

# G09/G16 MATLAB Toolbox

A parsing and visualisation toolbox for Gaussian 09/16 output files

Sebastiano Trusso

CNR – Istituto per i Processi Chimico-Fisici (IPCF), Messina, Italy

Version 1.0 — 2026

## Contents

|          |                                                               |           |
|----------|---------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                           | <b>3</b>  |
| <b>2</b> | <b>File I/O utilities</b>                                     | <b>4</b>  |
| 2.1      | G09_read_lines . . . . .                                      | 4         |
| 2.2      | G09_check_gaussian_match / G16_check_gaussian_match . . . . . | 4         |
| <b>3</b> | <b>Core data extraction</b>                                   | <b>5</b>  |
| 3.1      | G09_route / G16_route . . . . .                               | 5         |
| 3.2      | G09_structure / G16_structure . . . . .                       | 5         |
| 3.3      | G09_energy / G16_energy . . . . .                             | 6         |
| 3.4      | G09_charge_mult / G16_charge_mult . . . . .                   | 7         |
| 3.5      | G09_convergence / G16_convergence . . . . .                   | 7         |
| 3.6      | G09_dipole_polar / G16_dipole_polar . . . . .                 | 8         |
| 3.7      | G09_nmodes / G16_nmodes . . . . .                             | 8         |
| 3.8      | G09_spectra / G16_spectra . . . . .                           | 9         |
| <b>4</b> | <b>Atomic charges</b>                                         | <b>11</b> |
| 4.1      | G09_charges / G16_charges . . . . .                           | 11        |
| 4.2      | G09_charges_fchk (G09 only) . . . . .                         | 12        |
| <b>5</b> | <b>Formatted checkpoint (.fchk) support</b>                   | <b>13</b> |
| 5.1      | G09_fchk_read (G09 only) . . . . .                            | 13        |
| <b>6</b> | <b>Molecule and normal-mode visualisation</b>                 | <b>14</b> |
| 6.1      | G09_draw_molecule / G16_draw_molecule . . . . .               | 14        |
| 6.2      | G09_draw_mode / G16_draw_mode . . . . .                       | 15        |
| 6.3      | G09_modeViewer / G16_modeViewer . . . . .                     | 15        |
| 6.4      | G09_animate_mode / G16_animate_mode . . . . .                 | 16        |
| <b>7</b> | <b>Orbital energy analysis</b>                                | <b>18</b> |
| 7.1      | G09_orbital_energies / G16_orbital_energies . . . . .         | 18        |
| 7.2      | G09_draw_orbital / G16_draw_orbital . . . . .                 | 19        |
| <b>8</b> | <b>Response properties (G16 only)</b>                         | <b>20</b> |
| 8.1      | G16_hyperspherical . . . . .                                  | 20        |
| 8.2      | G16_tddft . . . . .                                           | 20        |

|           |                                                                                  |           |
|-----------|----------------------------------------------------------------------------------|-----------|
| <b>9</b>  | <b>Structural analysis utilities</b>                                             | <b>22</b> |
| 9.1       | G09_get_bond_length / G16_get_bond_length . . . . .                              | 22        |
| 9.2       | G09_gaussian_version / G16_gaussian_version . . . . .                            | 22        |
| 9.3       | G09_restart (G09 only) . . . . .                                                 | 23        |
| 9.4       | G_read_input (shared, no G09_/G16_ prefix) . . . . .                             | 23        |
| 9.5       | G09_list / G16_list . . . . .                                                    | 24        |
| <b>10</b> | <b>One-call data extraction</b>                                                  | <b>25</b> |
| 10.1      | G09_read_all / G16_read_all . . . . .                                            | 25        |
| <b>11</b> | <b>Report generation</b>                                                         | <b>26</b> |
| 11.1      | G09_write_report / G16_write_report . . . . .                                    | 26        |
| 11.2      | Function reference . . . . .                                                     | 27        |
| 11.3      | Selected signatures . . . . .                                                    | 28        |
| 11.4      | Notes on the port . . . . .                                                      | 29        |
| <b>12</b> | <b>Publication and licensing</b>                                                 | <b>32</b> |
| 12.1      | Declaration of Generative AI and AI-assisted technologies in the writing process | 32        |
| 12.2      | Note on the Use of AI-Assisted Development Tools . . . . .                       | 32        |
| 12.3      | Licence . . . . .                                                                | 32        |
| 12.4      | Citation . . . . .                                                               | 32        |

# 1 Introduction

This manual documents the **G09/G16 Toolbox**, a set of MATLAB<sup>®</sup> functions for parsing, analysing, and visualising Gaussian 09<sup>®</sup> and Gaussian 16<sup>®</sup> output files (`.out/.log`) and formatted checkpoint files (`.fchk`). The toolbox covers:

- extraction of molecular geometry, energies, thermochemistry, dipole moment, polarisability, and hyperpolarisability;
- vibrational analysis: normal modes, IR/Raman spectra, and interactive mode visualisation;
- molecular-orbital energy diagrams and TD-DFT excited-state analysis;
- 3D molecular structure rendering (CPK ball-and-stick models) with atomic charge and dipole overlays;
- utility functions for bond-length tables, Gaussian version detection, restart-file generation, and toolbox self-documentation.

Function names follow the `G09_xxx/G16_xxx` convention, indicating which Gaussian version's output format the function targets. Where the two versions behave identically, this manual documents them in a single combined subsection (`G09_xxx / G16_xxx`); where behaviour differs, the differences are called out explicitly.

Every parsing function that reads a `.out/.log` file accepts an optional `'Lines'` Name-Value parameter: a cell array of pre-read file lines. This lets `G09_read_all/G16_read_all` read the file from disk once and reuse the same lines across every sub-parser, instead of each function re-reading and re-splitting the file independently. Passing `'Lines'` is entirely optional — omitting it falls back to reading the file normally.

**Wrong-toolbox warning.** Because Gaussian 09 and Gaussian 16 output formats differ, using the wrong toolbox on a file (e.g. `G16_energy` on a Gaussian 09 file) can silently misparse it rather than raising an obvious error — for example, `G16_dipole_polar` on a Gaussian 09 file returns `NaN` for the polarisability instead of an error. To catch this, every function that reads a file from disk checks the Gaussian-version citation line (`"Gaussian NN, Revision X.YY,"`) it just read and prints a non-blocking `warning(...)` if `NN` does not match the toolbox being used, suggesting the other one instead. The check runs only once per actual disk read (see Section 2.2), so `G09_read_all/G16_read_all` produce a single warning, not one per sub-parser.

## 2 File I/O utilities

### 2.1 G09\_read\_lines

```
lines = G09_read_lines(filename)
```

Low-level reader used internally by every other **G09\_XXX** parser. Reads a Gaussian 09 output file and returns a cell array of lines, with line endings stripped. Transparently handles CRLF line endings (as produced by Windows G09W builds) and ISO-8859-1 (Latin-1) encoding.

G16 parsers read files directly with the platform default encoding rather than calling a separate **G16\_read\_lines** helper; the 'Lines' parameter interface is identical across both toolboxes.

### 2.2 G09\_check\_gaussian\_match / G16\_check\_gaussian\_match

```
ok = G09_check_gaussian_match(lines, filename)
ok = G16_check_gaussian_match(lines, filename)
```

Internal helper, called automatically by every function that reads a **.out/.log** file from disk (not part of the typical user-facing API, but documented here for completeness). Scans the already-read **lines** (no extra disk I/O) for the "**Gaussian NN, Revision X.YY,**" citation line printed near the top of every output file. If found and **NN** does not match the toolbox (9 for **G09\_\***, 16 for **G16\_\***), prints a non-blocking warning suggesting the other toolbox; otherwise does nothing. Returns **true** if no mismatch was detected (or the version is unknown), **false** if a warning was issued.

Because this check lives inside each function's own "read the file fresh from disk" branch, it only fires when a file is actually opened — if a function receives pre-read **lines** via the 'Lines' parameter (e.g. every sub-parser called by **G09\_read\_all/G16\_read\_all**), it skips both the disk read and the check, so a single call to **read\_all** produces exactly one warning, not one per sub-parser.

## 3 Core data extraction

### 3.1 G09\_route / G16\_route

```
route = G09_route(filename)
route = G16_route(filename)
route = G16_route(filename, 'Lines', lines)
```

Extracts the complete route section (the # line(s) between the two ---- separators) as a single string.

| Option  | Default | Description                                                 |
|---------|---------|-------------------------------------------------------------|
| 'Lines' | {}      | Pre-read file lines (Section 2.1); read normally if omitted |

```
r = G09_route('indaco.log');
% '# opt freq=raman b3lyp/6-311++g(d,p) nosymm geom=connectivity field
=x+50'
```

Both `G09_route/G16_route` and `G09_charge_mult/ G16_charge_mult` now accept the 'Lines' option, consistent with the rest of the toolbox.

### 3.2 G09\_structure / G16\_structure

```
mol = G09_structure(filename)
mol = G09_structure(filename, 'step', 'last')
mol = G09_structure(filename, 'step', 'first')
mol = G09_structure(filename, 'step', N)

mol = G16_structure(filename)
mol = G16_structure(filename, 'orientation', 'standard')
mol = G16_structure(filename, 'orientation', 'input')
```

Extracts the molecular geometry from a Gaussian output file.

**Difference:** Gaussian 09 writes only "Input orientation:" blocks (never "Standard orientation:"), so `G09_structure` has no 'orientation' option. Gaussian 16 can write either, so `G16_structure` exposes 'orientation' = 'standard' (default, falls back to input if absent) | 'input' | 'auto'.

| Option        | Default           | Description                                                           |
|---------------|-------------------|-----------------------------------------------------------------------|
| 'step'        | 'last'            | Which geometry block to extract: 'first', 'last', or an integer index |
| 'orientation' | 'auto' (G16 only) | 'standard'   'input'   'auto'                                         |
| 'Lines'       | {}                | Pre-read file lines; read normally if omitted                         |

| Field        | Type   | Description                                                      |
|--------------|--------|------------------------------------------------------------------|
| .symbols     | cell   | Atomic symbols                                                   |
| .xyz         | double | Coordinates in Å (X Y Z)                                         |
| .Z           | int    | Atomic numbers                                                   |
| .Natoms      | int    | Number of atoms                                                  |
| .step        | int    | Index of the extracted step (1-based)                            |
| .n_steps     | int    | Total geometry blocks found in the file                          |
| .orientation | char   | 'Input orientation' (always, G09) or<br>'standard'/'input' (G16) |
| .filename    | char   | Source file name                                                 |

```
mol = G09_structure('indaco.log');
mol = G16_structure('calc.out', 'orientation', 'input');
```

### 3.3 G09\_energy / G16\_energy

```
en = G09_energy(filename)
en = G09_energy(filename, 'step', 'last')
en = G09_energy(filename, 'step', N)
```

Extracts SCF energy and thermochemistry data.

| Field        | Type    | Description                                                                       |
|--------------|---------|-----------------------------------------------------------------------------------|
| .SCF         | double  | SCF energy (Hartree)                                                              |
| .method      | char    | Method string, e.g. 'RB3LYP'                                                      |
| .ZPE_corr    | double  | Zero-point correction (Hartree)                                                   |
| .U_corr      | double  | Thermal correction to energy (Hartree)                                            |
| .H_corr      | double  | Thermal correction to enthalpy (Hartree)                                          |
| .G_corr      | double  | Thermal correction to Gibbs free energy (Hartree)                                 |
| .E0          | double  | SCF + ZPE (Hartree)                                                               |
| .U           | double  | SCF + U_corr (Hartree, G16 only)                                                  |
| .H           | double  | SCF + H_corr (Hartree)                                                            |
| .G           | double  | SCF + G_corr (Hartree)                                                            |
| .S_JmolK     | double  | Entropy $S$ (J mol <sup>-1</sup> K <sup>-1</sup> , G09)                           |
| .ZPE_kJ      | double  | Zero-point energy (kJ/mol)                                                        |
| .T, .P       | double  | Temperature (K), pressure (atm)                                                   |
| .has_thermo  | logical | false for opt-only jobs (no freq); *_corr, E0, U, H, G, T, P are NaN in that case |
| .G_kJ, .H_kJ | double  | $G$ , $H$ in kJ/mol (G09)                                                         |
| .filename    | char    |                                                                                   |

| Option  | Default | Description                                                 |
|---------|---------|-------------------------------------------------------------|
| 'step'  | 'last'  | Which SCF block to extract: 'last', 'first', or integer $N$ |
| 'Lines' | {}      | Pre-read file lines                                         |

### 3.4 G09\_charge\_mult / G16\_charge\_mult

```
[charge, mult] = G09_charge_mult(filename)
[charge, mult] = G16_charge_mult(filename)
[charge, mult] = G16_charge_mult(filename, 'Lines', lines)
```

Extracts molecular charge and spin multiplicity from the "Charge = ... Multiplicity = ..." line. Format identical across G09 and G16.

```
[q, m] = G09_charge_mult('indaco.log'); % q = 0, m = 1
```

### 3.5 G09\_convergence / G16\_convergence

```
cv = G09_convergence(filename)
cv = G09_convergence(filename, 'plot', true)

cv = G16_convergence(filename)
cv = G16_convergence(filename, 'plot', true)
```

Extracts geometry-optimisation convergence criteria (Max/RMS Force, Max/RMS Displacement) at every optimisation step, together with their thresholds and whether/when the job converged. Block format identical across G09 and G16.

| Option | Default | Description                                               |
|--------|---------|-----------------------------------------------------------|
| 'plot' | false   | Plot the four convergence series against their thresholds |

| Field                                                             | Type    | Description                              |
|-------------------------------------------------------------------|---------|------------------------------------------|
| .MaxForce,<br>.RMSForce                                           | vector  | Max / RMS force per step (a.u.)          |
| .MaxDisp,<br>.RMSDisp                                             | vector  | Max / RMS displacement per step (a.u.)   |
| .thr_MaxForce,<br>.thr_RMSForce,<br>.thr_MaxDisp,<br>.thr_RMSDisp | double  | Corresponding convergence thresholds     |
| .converged                                                        | logical | Whether all four criteria were satisfied |
| .conv_step                                                        | int/NaN | Step at which convergence was reached    |
| .Nsteps                                                           | int     | Number of optimisation steps found       |
| .filename                                                         | char    |                                          |

```
cv = G16_convergence('V_E00t.out', 'plot', true);
cv.converged
cv.conv_step
```

### 3.6 G09\_dipole\_polar / G16\_dipole\_polar

```
dp = G09_dipole_polar(filename)
dp = G09_dipole_polar(filename, 'units', 'Debye')

dp = G16_dipole_polar(filename)
dp = G16_dipole_polar(filename, 'units', 'SI')
```

Extracts the dipole moment and polarisability tensor.

**Differences between G09 and G16:** G09 prints "Exact polarizability:" as six values on a single line (upper triangle, row order), rather than G16's formatted Alpha(0;0)/Alpha(-w;w) blocks. G09 has no dynamic (laser-frequency-dependent) polarisability unless requested via Polar=OptRot or similar keywords, and additionally prints a "Diagonal vibrational polarizability:" line (3 values, returned in .alpha\_vib). An approximate CPHF tensor, if printed by G09 ("Approx polarizability:"), is returned in .alpha\_approx. G16 instead reports full dynamic polarisability entries per laser frequency (.alpha\_dyn) and modal derivatives .dmu\_dQ/.dalpha\_dQ (related to IR intensities and Raman activities; for exact scaled values use G16\_spectra instead).

| Option  | Default | Description                                                    |
|---------|---------|----------------------------------------------------------------|
| 'units' | 'au'    | 'au'   'Debye' (dipole only, G09) / 'Debye' (full, G16)   'SI' |
| 'Lines' | {}      | Pre-read file lines                                            |

| Field                                   | Description                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------|
| .mu_x, .mu_y, .mu_z, .mu_tot, .mu_units | Dipole components, magnitude, unit string                                         |
| .alpha_iso, .alpha_aniso                | Isotropic polarisability, anisotropy                                              |
| .alpha_tensor, .alpha_units             | $3 \times 3$ static tensor (a.u.), unit string                                    |
| .alpha_vib, .has_alpha_vib              | Diagonal vibrational polarisability (a.u.), logical (G09)                         |
| .alpha_approx                           | CPHF approximate tensor (a.u.), NaN if absent (G09)                               |
| .alpha_dyn, .N_dyn                      | Struct array of dynamic polarisability entries (per laser frequency), count (G16) |
| .dmu_dQ, .dalpha_dQ, .has_deriv         | Dipole/polarisability derivatives w.r.t. normal modes, logical (G16)              |
| .filename                               |                                                                                   |

### 3.7 G09\_nmodes / G16\_nmodes

```
nm = G09_nmodes(filename)
nm = G09_nmodes(filename, 'modes', [5 10 20])

nm = G16_nmodes(filename)
```



```
nm = G16_nmodes(filename, 'section', 'last')
```

Extracts normal-mode displacement vectors.

G09 writes only one *Harmonic frequencies* section per file (even for opt+freq jobs), so `G09_nmodes` has no `'section'` option — it is implicit. G16 may contain more than one such section (e.g. a preliminary and a final one) and defaults to the last.

| Option    | Default           | Description                              |
|-----------|-------------------|------------------------------------------|
| 'modes'   | []                | Restrict output to specific mode indices |
| 'section' | 'last' (G16 only) | 'last'   'first'                         |
| 'Lines'   | {}                | Pre-read file lines                      |

  

| Field       | Description                                                                          |
|-------------|--------------------------------------------------------------------------------------|
| .freq,      | .IR, Frequencies ( $\text{cm}^{-1}$ ), IR intensities (KM/Mole), Raman activities    |
| .Raman      | ( $\text{\AA}^4/\text{AMU}$ , [] if absent)                                          |
| .redmass,   | Reduced masses (AMU), force constants ( $\text{mDyne/\AA}$ )                         |
| .frconst    |                                                                                      |
| .symmetry   | Symmetry labels                                                                      |
| .disp       | $[N_{\text{atoms}} \times 3 \times N_{\text{modes}}]$ Cartesian displacement vectors |
| .Nmodes,    |                                                                                      |
| .Natoms,    |                                                                                      |
| .has_Raman, |                                                                                      |
| .filename   |                                                                                      |

### 3.8 G09\_spectra / G16\_spectra

```
sp = G09_spectra(filename)
sp = G09_spectra(filename, Name, Value, ...)

sp = G16_spectra(filename)
sp = G16_spectra(filename, 'section', 'last')
```

Extracts IR/Raman line spectra and generates Lorentzian-broadened continuous spectra.

G09 writes only one Harmonic-frequencies section per file, so `G09_spectra` needs no `'section'` parameter; `G16_spectra` exposes it to select which section to use when more than one is present.

| Option      | Default           | Description                                 |
|-------------|-------------------|---------------------------------------------|
| 'FWHM'      | 10                | Lorentzian FWHM ( $\text{cm}^{-1}$ )        |
| 'xmin'      | 0                 | Lower wavenumber limit ( $\text{cm}^{-1}$ ) |
| 'xmax'      | 4000              | Upper wavenumber limit ( $\text{cm}^{-1}$ ) |
| 'dx'        | 1                 | Grid step ( $\text{cm}^{-1}$ )              |
| 'normalize' | false             | Normalise continua to maximum = 1           |
| 'plot'      | false             | Generate figure                             |
| 'section'   | 'last' (G16 only) | 'last'   'first'                            |
| 'Lines'     | {}                | Pre-read file lines                         |

| Field                  | Description                                                                            |
|------------------------|----------------------------------------------------------------------------------------|
| .freq,<br>.Raman       | .IR, Frequencies ( $\text{cm}^{-1}$ ), IR intensities, Raman activities ([] if absent) |
| .has_Raman,<br>.Nmodes | Logical, number of modes                                                               |
| .x,<br>.Raman_cont     | .IR_cont, Wavenumber grid, broadened IR/Raman continua                                 |
| .FWHM,<br>.filename    |                                                                                        |

```
sp = G16_spectra('calc.out', 'FWHM', 15, 'xmin', 400, 'plot', true);
```

## 4 Atomic charges

### 4.1 G09\_charges / G16\_charges

```
ch = G09_charges(filename)
ch = G09_charges(filename, 'type', 'APT')
ch = G09_charges(filename, 'mode', 'heavy')
ch = G09_charges(filename, 'plot', true, 'threshold', 0.05)
```

Extracts Mulliken or APT atomic charges and optionally renders them on the 3D structure, with an optional dipole-moment arrow overlay.

**Robustness:** handles all known Mulliken label variants across Gaussian revisions/OSes (e.g. "Mulliken atomic charges:" on G09 Rev. A.02/Windows and some Linux builds; "Mulliken charges:" on G09 Rev. B.01/C.01/D.01 and on all G16 revisions), by matching any line containing the charge-type keyword and "charges:", excluding "Sum of" and H-summed lines. The exact label found is reported in `ch.label`.

| Option            | Default     | Description                                                                                   |
|-------------------|-------------|-----------------------------------------------------------------------------------------------|
| 'type'            | 'Mulliken'  | 'Mulliken'   'APT'                                                                            |
| 'mode'            | 'atom'      | 'atom'   'heavy' (H charges summed onto heavy atoms)                                          |
| 'plot'            | true        | Render labels on the 3D structure                                                             |
| 'AtomScale'       | 0.35        | CPK sphere scale                                                                              |
| 'BondTol'         | 1.30        | Bond detection tolerance                                                                      |
| 'FontSize'        | 8           | Charge-label font size                                                                        |
| 'ColorScale'      | 'RdBu'      | 'RdBu' (blue=neg/red=pos)   'none'                                                            |
| 'threshold'       | 0           | Hide labels with $ q  < \text{threshold}$                                                     |
| 'ShowDipole'      | false       | Overlay the total dipole moment vector                                                        |
| 'DipoleOrigin'    | 'negcharge' | Arrow anchor: 'negcharge' (centroid of negative charges)   'poscharge'   'centroid'   [x y z] |
| 'DipoleScale'     | 1.0         | Arrow length, Å per Debye                                                                     |
| 'DipoleColor'     | [0 0.6 0]   | Arrow colour                                                                                  |
| 'DipoleLineWidth' | 2.5         | Arrow line width                                                                              |
| 'ShowDipoleLabel' | true        | Annotate the arrow with $ \mu $                                                               |
| 'DipoleFontSize'  | 11          | Dipole label font size                                                                        |
| 'DipoleUnits'     | 'Debye'     | Units for $ \mu $ : 'Debye'   'au'                                                            |
| 'Lines'           | {}          | Pre-read file lines                                                                           |

| Field                 | Description                                             |
|-----------------------|---------------------------------------------------------|
| .symbols,<br>.charges | Atomic symbols, per-atom charges (e)                    |
| .charges_H            | H-summed charges on heavy atoms                         |
| .sum_q                | Sum of all charges ( $\approx 0$ for neutral molecules) |
| .type, .label         | 'Mulliken'/'APT', exact header label found              |
| .Natoms               |                                                         |

## 4.2 G09\_charges\_fchk (G09 only)

```
ch_out = G09_charges_fchk(mol, ch)
ch_out = G09_charges_fchk(mol, ch, Name, Value, ...)
```

Visualises atomic charges directly from a `G09_fchk_read` struct, without requiring the corresponding `.log/.out` file. Mirrors the interface of `G09_charges` exactly, so the two can be used interchangeably once `data.mol` and `data.ch` are available.

| Option       | Default | Description                                                                                                                                                                             |
|--------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 'mode'       | 'atom'  | 'atom'   'heavy' (requires <code>ch.charges_H</code> non-empty — <code>.fchk</code> files have no native H-summed data; use <code>G09_charges</code> on the <code>.log</code> for that) |
| 'plot'       | true    |                                                                                                                                                                                         |
| 'AtomScale'  | 0.35    |                                                                                                                                                                                         |
| 'BondTol'    | 1.30    |                                                                                                                                                                                         |
| 'FontSize'   | 8       |                                                                                                                                                                                         |
| 'ColorScale' | 'RdBu'  | 'RdBu'   'none'                                                                                                                                                                         |
| 'threshold'  | 0       |                                                                                                                                                                                         |

```
data = G09_fchk_read('3typ.fchk');
ch    = G09_charges_fchk(data.mol, data.ch); %
      default
ch    = G09_charges_fchk(data.mol, data.ch, 'plot', false, 'threshold',
      0.05);
ch    = G09_charges_fchk(data.mol, data.ch, 'ColorScale', 'none'); %
      drop-in for G09_charges
```

Requires `mol` with fields `.symbols`, `.xyz`, `.Natoms`, `.filename`, and `ch` with fields `.charges`, `.symbols`, `.type`, `.Natoms` (`.charges_H` optional).

No equivalent `G16_charges_fchk` has been provided yet: since `.fchk` format is identical for G09 and G16, `G09_charges_fchk` can be used with `.fchk` files from either program version.

## 5 Formatted checkpoint (.fchk) support

### 5.1 G09\_fchk\_read (G09 only)

```
data = G09_fchk_read(filename)
data = G09_fchk_read(filename, 'verbose', false)
```

Reads a Gaussian 09/16 formatted checkpoint file (.fchk), generated with `formchk jobname.chk jobname.fchk`. Uses a single generic section parser (no hard-coded line counts) and is compatible with both G09 and G16 .fchk files.

Supersedes the old .fck/newzmat-format reader: reads the modern .fchk ASCII format, returns SI-ready fields (XYZ in Å, dipole in Debye), and needs no external helper functions. Despite the G09\_ prefix, this function works on .fchk files produced by *either* G09 or G16, since the formatted-checkpoint layout does not depend on the Gaussian version.

| Option    | Default | Description                           |
|-----------|---------|---------------------------------------|
| 'verbose' | true    | Print progress messages while parsing |

**Output struct data** (grouped by section):

| Field                                     | Description                                                      |
|-------------------------------------------|------------------------------------------------------------------|
| .title, .method, .basis                   | First line, method (e.g. 'RB3LYP'), basis set (e.g. 'def2SVPP')  |
| .Nat, .charge, .mult                      | Number of atoms, molecular charge, spin multiplicity             |
| .Nelec, .Nalpha, .Nbeta                   | Total, alpha, beta electron counts                               |
| .Nbasis, .Nbasis_indep                    | Basis functions, independent basis functions                     |
| .symbols, .AN, .masses                    | Atomic symbols, atomic numbers, real atomic masses (AMU)         |
| .xyz, .xyz_bohr                           | Coordinates (Å), coordinates (Bohr)                              |
| .SCF_energy, .total_energy, .virial_ratio | SCF energy, total energy (Hartree), virial ratio ( $\approx 2$ ) |
| .alpha_orb_energies, .beta_orb_energies   | MO energies (Hartree); beta is [] if RHF                         |
| .alpha_MO_coeff                           | Alpha MO coefficients (column-major)                             |
| .mulliken_charges                         | Mulliken charges (e)                                             |
| .HOMO_idx, .HOMO_eV, .LUMO_eV, .gap_eV    | Frontier-orbital summary                                         |
| .gradient, .rms_force                     | Cartesian gradient (Hartree/Bohr), RMS force                     |

## 6 Molecule and normal-mode visualisation

### 6.1 G09\_draw\_molecule / G16\_draw\_molecule

```
G09_draw_molecule(mol)
G09_draw_molecule(mol, 'Name', Value, ...)
```

Renders a 3D CPK ball-and-stick model from a mol struct (as returned by G09\_structure/G16\_structure), with bonds detected automatically from a covalent-radius criterion and drawn as 1, 2, or 3 parallel lines according to an estimated bond order.

| Option       | Default          | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 'AtomScale'  | 0.35             | Sphere radius as a fraction of the covalent radius                                                                                                                                                                                                                                                                                                                                                         |
| 'BondTol'    | 1.30             | Bond tolerance: bonded if $d < (r_1 + r_2) \times \text{BondTol}$                                                                                                                                                                                                                                                                                                                                          |
| 'ShowLabels' | true             | Show atom labels ("C1", "H2", ...)                                                                                                                                                                                                                                                                                                                                                                         |
| 'ShowLegend' | true             | Show the element-colour legend                                                                                                                                                                                                                                                                                                                                                                             |
| 'Title'      | filename         | Figure title                                                                                                                                                                                                                                                                                                                                                                                               |
| 'BgColor'    | [0.95 0.95 0.95] | Axes background colour                                                                                                                                                                                                                                                                                                                                                                                     |
| 'Ax'         | (new figure)     | Draw into an existing axes handle                                                                                                                                                                                                                                                                                                                                                                          |
| 'ShowAxes'   | false            | Draw a small Cartesian X/Y/Z reference triad, anchored at the lower-left-front corner of the plotted molecule                                                                                                                                                                                                                                                                                              |
| 'AxesLength' | []               | Length of the X/Y/Z arrows, in Å ([] = auto, 20% of the molecule's bounding-box diagonal)                                                                                                                                                                                                                                                                                                                  |
| 'BondList'   | []               | [ $N_{\text{bonds}} \times 2$ ] or [ $N_{\text{bonds}} \times 3$ ] explicit atom-index pairs (optionally with a 3rd column giving a pre-computed bond order 1/2/3) to draw as bonds, bypassing the BondTol distance criterion ([] = auto-detect). Used by G09_animate_mode/G16_animate_mode to keep a fixed bond topology (and, with the 3rd column, a fixed bond order) across all frames of an animation |

```
mol = G09_structure('zeatin.out');
G09_draw_molecule(mol);
G09_draw_molecule(mol, 'ShowLabels', false, 'AtomScale', 0.4);
G09_draw_molecule(mol, 'ShowAxes', true);
```

**Bond order (single/double/triple).** For C-C, C-N, and C-O pairs, bond order is estimated purely from bond length (any other element pair is always drawn as a single bond) and rendered as 1/2/3 parallel lines in the usual chemical-drawing convention — a geometric estimate, not Gaussian's own bond-order analysis (e.g. Wiberg/NBO indices). The C-C double/single threshold is deliberately set to 1.36 Å rather than the generic reference-length midpoint ( $\approx 1.44$  Å): symmetric aromatic rings have essentially equal C-C bond lengths ( $\approx 1.39$ – $1.40$  Å, with no real single/double alternation to recover from geometry alone), so the generic midpoint would classify every ring bond as double. With the adjusted threshold, aromatic rings are instead rendered as all-single — a more honest depiction than an arbitrary alternating pattern, since the true bond order is  $\approx 1.5$  all the way around the ring.

No differences between the G09 and G16 versions beyond the input struct's origin.

## 6.2 G09\_draw\_mode / G16\_draw\_mode

```
G09_draw_mode(mol, nm, mode_idx)
G09_draw_mode(mol, nm, mode_idx, 'Name', Value, ...)
```

Renders a single vibrational normal mode as displacement arrows over the CPK structure.

| Option       | Default     | Description                 |
|--------------|-------------|-----------------------------|
| 'Scale'      | 1.5         | Arrow length scale          |
| 'ArrowColor' | [1 0.4 0.1] | Arrow colour                |
| 'AtomScale'  | 0.35        | CPK sphere scale factor     |
| 'BondTol'    | 1.30        | Bond detection tolerance    |
| 'ShowLabels' | false       | Show atom index labels      |
| 'FlipSign'   | false       | Invert all arrow directions |

**Why 'FlipSign' exists.** Normal-mode eigenvectors have an arbitrary overall sign — a mode and its 180°-phase-shifted twin are physically identical. If the arrows point opposite to GaussView's rendering for a given mode, set 'FlipSign', **true** to flip them. This is purely cosmetic and has no effect on frequencies or intensities.

## 6.3 G09\_modeViewer / G16\_modeViewer

```
G09_modeViewer(filename)
G09_modeViewer(filename, 'Name', Value, ...)
```

Interactive selector GUI for browsing vibrational modes. Reads the structure and all normal modes via `G09.structure/G09.nmodes`, then opens a window listing every mode (index, frequency, symmetry). Selecting an entry calls `G09_draw_mode` to render that mode in a new figure window; previously drawn mode figures stay open, so different modes (or render settings) can be compared side by side.

### Window controls:

- *Order by:* sort the mode list by mode number (default), IR intensity, or Raman intensity (descending; only offered when Raman data is present). The current selection is preserved across reordering.
- A text box to set a custom title on the current mode figure.
- A control for every `G09_draw_mode` option (`Scale`, `ArrowColor`, `AtomScale`, `BondTol`, `ShowLabels`, `FlipSign`); changing any of them immediately redraws the selected mode in a new figure. Arrow colour can be set from a named preset (Orange/Red/Blue/Green/Black/ Magenta) or a custom colour.
- A *Save figure...* button exporting the target mode figure to JPEG, EPS, or PDF (EPS/PDF as true vector graphics).
- An *Animate mode (MP4)...* button exporting an MP4 animation of the currently selected mode via `G09_animate_mode/ G16_animate_mode`, using the current `Scale/AtomScale/ BondTol/ShowLabels/FlipSign` settings and, automatically, the camera orientation of the currently displayed mode figure (a manual rotation carries over into the animation).

Name-Value arguments passed to `G09_modeViewer` pre-populate the corresponding option control(s), e.g. `G09_modeViewer('file.log', 'Scale', 2, 'ShowLabels', true)`.

```
G09_modeViewer('V_E00t.out');
G16_modeViewer('INDACO_E025_ppDP.LOG', 'Scale', 2, 'ShowLabels', true)
;
```

See also `G09_structure/G16_structure`, `G09_nmodes/G16_nmodes`, `G09_draw_mode/G16_draw_mode`, `G09_animate_mode/G16_animate_mode`.

## 6.4 G09\_animate\_mode / G16\_animate\_mode

```
outfile = G09_animate_mode(mol, nm, mode_idx)
outfile = G09_animate_mode(mol, nm, mode_idx, 'Name', Value, ...)

outfile = G16_animate_mode(mol, nm, mode_idx)
```

Exports an MP4 animation of a single vibrational mode: the molecule oscillates along the mode's displacement vector (equilibrium  $\pm$  amplitude), in the style of GaussView's mode animations, and the result is saved via `VideoWriter` ('MPEG-4' profile). A figure window opens and animates live while the video is rendered; avoid interacting with it until the function returns.

| Option           | Default   | Description                                                                                                                                                                                                              |
|------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 'Filename'       | [] (auto) | Output path; default <source>.mode<N>.mp4, .mp4 appended if missing                                                                                                                                                      |
| 'Scale'          | 1.5       | Displacement amplitude scale, same meaning as <code>G09_draw_mode/G16_draw_mode</code> 's 'Scale'                                                                                                                        |
| 'FlipSign'       | false     | Invert the displacement direction                                                                                                                                                                                        |
| 'AtomScale'      | 0.35      | See <code>G09_draw_molecule/G16_draw_molecule</code>                                                                                                                                                                     |
| 'BondTol'        | 1.30      | See <code>G09_draw_molecule/G16_draw_molecule</code> ; also used to compute the fixed bond list (see below)                                                                                                              |
| 'ShowLabels'     | false     | See <code>G09_draw_molecule/G16_draw_molecule</code>                                                                                                                                                                     |
| 'FramesPerCycle' | 30        | Frames per oscillation period                                                                                                                                                                                            |
| 'NCycles'        | 2         | Number of periods rendered                                                                                                                                                                                               |
| 'FPS'            | 20        | Video frame rate                                                                                                                                                                                                         |
| 'View'           | []        | [azimuth elevation] in degrees, the starting camera orientation ([] = MATLAB's default 3D view, azimuth= -37.5, elevation= 30). Pass the output of MATLAB's own <code>view(ax)</code> to match an already-rotated figure |
| 'ShowWaitbar'    | true      | Print rendering progress to the command window                                                                                                                                                                           |

```
mol = G16_structure('V_E00t.out');
nm = G16_nmodes('V_E00t.out');
G16_animate_mode(mol, nm, 70, 'Scale', 2, 'FPS', 25);
```

**Fixed bond list and bond order.** Bonds (and their estimated order, 1/2/3) are computed once from the *equilibrium* geometry (via `G09_get_bond_length/G16_get_bond_length`, using 'BondTol' as the tolerance) and passed to `G09_draw_molecule/G16_draw_molecule` through its 'BondList' parameter (atom-index pairs plus a 3rd bond-order column), instead of being re-detected from



instantaneous inter-atomic distances on every frame. Without this, bonds would flicker in and out — and double/triple bonds would flicker to/from single — as oscillating atoms cross the relevant distance thresholds.

**No graphical waitbar.** Rendering progress is printed to the command window rather than shown with MATLAB's `waitbar`: on some MATLAB versions, `waitbar` itself was found to error intermittently across many rapid updates ("Not enough input arguments" inside its own internal `title()` call).

**Axes handling.** Each frame's axes are deleted and recreated (rather than cleared with `cla`) before redrawing: in this 3D-plus-lighting configuration, `cla(ax)` was found to leave stale graphics objects behind, causing the exported video to show all frames superimposed instead of one at a time.

No differences between the G09 and G16 versions beyond the input structs' origin.

See also `G09_draw_molecule/G16_draw_molecule`, `G09_draw_mode/G16_draw_mode`, `G09_modeViewer/G16_modeViewer`, `G09_get_bond_length/G16_get_bond_length`.

## 7 Orbital energy analysis

### 7.1 G09\_orbital\_energies / G16\_orbital\_energies

```
oe = G09_orbital_energies(filename)
oe = G09_orbital_energies(filename, 'step', 'last')
oe = G09_orbital_energies(filename, 'step', N)
```

Extracts molecular orbital energies (HOMO/LUMO and the full occupied/virtual spectrum) from a Gaussian 09/16 output file, by parsing the "Alpha occ. eigenvalues --" / "Alpha virt. eigenvalues --" lines printed by Gaussian's population analysis (and the corresponding "Beta" lines for open-shell calculations). These blocks repeat once per SCF calculation in the file (e.g. once per optimisation step); the 'step' option selects which block to report, consistently with G09\_energy/G09\_structure.

| Option | Default | Description                                                         |
|--------|---------|---------------------------------------------------------------------|
| 'step' | 'last'  | Which eigenvalue block to use: 'first', 'last', or an integer index |

**Output struct** oe (energies in Hartree unless noted):

| Field                         | Type    | Description                                          |
|-------------------------------|---------|------------------------------------------------------|
| .alpha_occ                    | vector  | Occupied alpha orbital energies                      |
| .alpha_virt                   | vector  | Virtual alpha orbital energies                       |
| .beta_occ                     | vector  | Occupied beta orbital energies ( [] if closed-shell) |
| .beta_virt                    | vector  | Virtual beta orbital energies ( [] if closed-shell)  |
| .has_beta                     | logical | <b>true</b> for open-shell (UHF/UKS) calculations    |
| .HOMO / .LUMO                 | double  | Highest occupied / lowest unoccupied orbital energy  |
| .gap                          | double  | LUMO -- HOMO                                         |
| .HOMO_alpha / .LUMO_alpha     | double  | Alpha HOMO/LUMO (== .HOMO/.LUMO for closed-shell)    |
| .HOMO_beta / .LUMO_beta       | double  | Beta HOMO/LUMO ( [] if closed-shell)                 |
| .HOMO_eV / .LUMO_eV / .gap_eV | double  | Same three quantities, in eV                         |
| .step                         | int     | Block index actually used                            |
| .Nsteps                       | int     | Number of eigenvalue blocks found in the file        |
| .filename                     | char    | Source file name                                     |

```
oe = G09_orbital_energies('V_E00t.out');
fprintf('HOMO-LUMO gap = %.3f eV\n', oe.gap_eV);
```

For open-shell (UHF/UKS) calculations, HOMO and LUMO are computed as  $\max(\text{HOMO}_\alpha, \text{HOMO}_\beta)$  and  $\min(\text{LUMO}_\alpha, \text{LUMO}_\beta)$  respectively, so .gap always refers to the smallest alpha/beta HOMO-LUMO gap. No differences between the G09 and G16 versions beyond the parsed file's origin.

## 7.2 G09\_draw\_orbital / G16\_draw\_orbital

```
G09_draw_orbital(oe)
G09_draw_orbital(oe, 'Name', Value, ...)
```

Draws a molecular-orbital energy-level diagram from the struct returned by `G09_orbital_energies/G16_orbital_energies` showing a user-selected number of occupied/virtual levels around the frontier orbitals and highlighting the HOMO–LUMO transition with an arrow labelled with the gap value.

| Option             | Default                              | Description                                                                                   |
|--------------------|--------------------------------------|-----------------------------------------------------------------------------------------------|
| 'NLevels'          | 5                                    | Number of occupied/virtual levels shown on each side of the frontier orbitals                 |
| 'Units'            | 'eV'                                 | Energy units: 'eV' or 'Hartree'                                                               |
| 'Ax'               | (new figure)                         | Draw into an existing axes handle                                                             |
| 'OccColor'         | [0.25 0.25 0.70]                     | Colour for non-frontier occupied levels                                                       |
| 'VirtColor'        | [0.70 0.30 0.25]                     | Colour for non-frontier virtual levels                                                        |
| 'HOMOColor'        | [0.00 0.45 0.85]                     | Highlight colour for the HOMO                                                                 |
| 'LUMOCOLOR'        | [0.90 0.35 0.00]                     | Highlight colour for the LUMO                                                                 |
| 'ArrowColor'       | [0.15 0.60 0.15]                     | HOMO→LUMO transition arrow colour                                                             |
| 'ShowLabels'       | true                                 | Annotate each level with its energy value                                                     |
| 'ShowSpins'        | true                                 | Draw a paired-electron arrow glyph ( $\uparrow\downarrow$ ) on occupied levels                |
| 'FontSize'         | 8                                    | Font size for level energy-value labels                                                       |
| 'FrontierFontSize' | 13                                   | Font size for the HOMO/LUMO tags and the gap annotation                                       |
| 'ArrowHeadSize'    | 0.2                                  | HOMO–LUMO arrow head size, as a fraction of the arrow length (passed to <code>quiver</code> ) |
| 'Title'            | filename or 'Orbital Energy Diagram' | Figure title                                                                                  |

```
oe = G09_orbital_energies('a1.out');
G09_draw_orbital(oe)
G09_draw_orbital(oe, 'NLevels', 8, 'Units', 'Hartree')
```

Requires the `oe` struct returned by `G09_orbital_energies/ G16_orbital_energies` and errors if `alpha_occ/alpha_virt` are missing or empty. No differences between the G09 and G16 versions beyond the input struct's origin.

## 8 Response properties (G16 only)

### 8.1 G16\_hyperpolar

```
hp = G16_hyperpolar(filename)
hp = G16_hyperpolar(filename, 'units', 'au')      % default
hp = G16_hyperpolar(filename, 'units', 'esu')    % 10-30 esu
hp = G16_hyperpolar(filename, 'units', 'SI')     % 10-50 C3 m3 J-2
```

Extracts the dipole hyperpolarisability  $\beta$  (static, dynamic, and vibrational contributions) from a Gaussian 16 output file.

| Option  | Default | Description                                                                                                    |
|---------|---------|----------------------------------------------------------------------------------------------------------------|
| 'units' | 'au'    | 'au'   'esu' (10 <sup>-30</sup> esu)   'SI' (10 <sup>-50</sup> C <sup>3</sup> m <sup>3</sup> J <sup>-2</sup> ) |

| Field             | Description                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------|
| .beta0.par_z,     | Parallel / perpendicular component of static $\beta(0;0,0)$ along $z$                                    |
| .beta0.perp_z     |                                                                                                          |
| .beta0.vec_x/y/z, | Vector components and magnitude $ \beta_{\text{vec}} $                                                   |
| .beta0.beta_vec   |                                                                                                          |
| .beta0.tensor     | All Cartesian tensor components ( $xxx, xyx, \dots, zzz$ )                                               |
| .beta_dyn         | Struct array, one entry per laser frequency: .lambda_nm, .par_z, .perp_z, .vec_x/y/z, .beta_vec, .tensor |
| .N_dyn            | Number of dynamic (laser) frequencies found                                                              |
| .beta_vib,        | Diagonal vibrational hyperpolarisability, logical                                                        |
| .has_vib          |                                                                                                          |
| .filename         |                                                                                                          |

```
hp = G16_hyperpolar('V_E00t.out');
hp.beta0.beta_vec          % static beta modulus, a.u.
hp.beta_dyn(1).lambda_nm  % wavelength of the first dynamic entry (nm)
hp.beta_dyn(1).beta_vec   % |beta| at that frequency
```

### 8.2 G16\_tddft

```
td = G16_tddft(filename)
td = G16_tddft(filename, 'nstates', N)      % first N states only
td = G16_tddft(filename, 'plot', true)      % also generate UV-Vis plot
td = G16_tddft(filename, 'FWHM_eV', 0.3)    % broadening FWHM (eV)
```

Extracts TD-DFT excited states and generates a Gaussian-broadened UV-Vis spectrum.

| Option    | Default | Description                            |
|-----------|---------|----------------------------------------|
| 'nstates' | Inf     | Load only the first $N$ excited states |
| 'plot'    | false   | Generate the UV-Vis plot               |
| 'FWHM_eV' | 0.30    | Gaussian broadening FWHM (eV)          |

| Field                  | Description                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| .n, .mult              | State index; multiplicity label (e.g. 'Singlet-A', 'Triplet-B')                                                                       |
| .eV, .nm, .f           | Excitation energy (eV), wavelength (nm), oscillator strength                                                                          |
| .S2, .has_S2           | $\langle S^2 \rangle$ expectation value (0 for pure singlets), logical                                                                |
| .trans                 | Cell array, one $[N_{\text{contrib}} \times 3]$ matrix per state: columns [MO_from, MO_to, coeff] (MO_to < 0 denotes a de-excitation) |
| .Nstates               | Total number of excited states loaded                                                                                                 |
| .x_nm, .eps_cont       | Wavelength grid (100–1000 nm), Gaussian-broadened UV-Vis spectrum (arb. units)                                                        |
| .FWHM_eV,<br>.filename |                                                                                                                                       |

```

td = G16_tddft('violacein_td.out');
plot(td.x_nm, td.eps_cont)      % UV-Vis spectrum
stem(td.nm, td.f)              % transition stem plot
td.trans{2}                    % transitions contributing to the 2nd
    state

```

No G09 equivalent is provided: TD-DFT excited-state analysis and hyperpolarisability response properties in this toolbox are currently supported only for Gaussian 16 output files.

## 9 Structural analysis utilities

### 9.1 G09\_get\_bond\_length / G16\_get\_bond\_length

```
bondTable = G09_get_bond_length(mol)
bondTable = G09_get_bond_length(mol, 'Name', Value, ...)
```

Builds the bond-length table of a molecule from a `mol` struct (fields `.symbols`, `.xyz`), detecting bonds via a covalent-radius criterion, and returns a MATLAB table.

| Option      | Default    | Description                                                                  |
|-------------|------------|------------------------------------------------------------------------------|
| 'Tolerance' | 1.15       | Multiplicative factor on the sum of covalent radii for the bonding criterion |
| 'IncludeH'  | true       | Include bonds involving hydrogen atoms                                       |
| 'SortBy'    | 'distance' | Sort by 'distance' (ascending) or 'atom' (atom index)                        |
| 'SaveAs'    | ''         | Path (.xlsx or .csv) to save the table to; not saved if omitted              |

Output columns: Atom1, Sym1, Atom2, Sym2, Distance Ang.

```
T = G09_get_bond_length(mol, 'SortBy', 'atom');
T = G16_get_bond_length(mol, 'IncludeH', false, 'SaveAs', 'bonds.xlsx');
```

The bonding criterion is purely geometric (covalent radii), not derived from Gaussian's own connectivity/topology matrix — suitable for quick structural analysis, not as the "official" G09/G16 connectivity.

### 9.2 G09\_gaussian\_version / G16\_gaussian\_version

```
gv = G09_gaussian_version(filename)
gv = G16_gaussian_version(filename)
```

Detects which Gaussian version/revision produced a `.out/.log/.fchk` file.

For `.out/.log` files, reads the "Gaussian NN, Revision X.YY," citation line printed near the top of the file (works for any NN: 09, 16, ...). `.fchk` files do not store this information at all; in that case the function looks for a sibling `.log/.out` file with the same base name in the same folder and reads the version from there. If no sibling file is found, `.major/.revision/.full` are returned empty and a warning is issued.

| Field                  | Description                                                 |
|------------------------|-------------------------------------------------------------|
| <code>.major</code>    | Gaussian major version, e.g. 9, 16 ([] if unknown)          |
| <code>.revision</code> | Revision string, e.g. 'A.02', 'C.01' ('' if unknown)        |
| <code>.full</code>     | Citation line as printed, e.g. 'Gaussian 09, Revision A.02' |
| <code>.source</code>   | 'out/log'   'fchk-sibling:<path>'   'unknown'               |
| <code>.filename</code> |                                                             |

```
gv = G09_gaussian_version('a1.out');          % gv.major = 16, gv.
revision = 'C.01'
```

```
gv = G16_gaussian_version('molecule.fchk'); % falls back to molecule.
out/.log if present
```

### 9.3 G09\_restart (G09 only)

```
gjf_file = G09_restart(filename)
gjf_file = G09_restart(filename, 'Name', Value, ...)
```

Generates a Gaussian 09/16 input file (.gjf) to restart a calculation from a geometry found in an existing .log/.out file.

| Option        | Default              | Description                                                          |
|---------------|----------------------|----------------------------------------------------------------------|
| 'step'        | 'last'               | 'last'   'first'   integer $N$ — which geometry step to restart from |
| 'output'      | <base>_restarted.gjf | Output .gjf path                                                     |
| 'extra_route' | ''                   | Additional keywords appended to the route section                    |
| 'nproc'       | 0                    | %nprocshared override (0 = keep/omit)                                |
| 'mem'         | ''                   | %mem override ('' = keep/omit)                                       |

Gaussian 09 Windows output files often lack a %chk line; in that case the checkpoint name is derived from the source file name. No G16\_restart has been provided yet; the generated .gjf is valid for restarting under either Gaussian 09 or Gaussian 16.

Returns the path of the generated .gjf file.

### 9.4 G\_read\_input (shared, no G09\_/G16\_ prefix)

```
ginp = G_read_input(filename)
```

Reads a Gaussian *input* file (.gjf, .com, or .in) and extracts the link0 commands, route section, title, charge/multiplicity, and starting Cartesian geometry.

The Gaussian input file format does not differ between Gaussian 09 and Gaussian 16, so this is the toolbox's only function without a G09\_/G16\_ prefix pair: an identical copy is kept in both the G09/ and G16/ folders, so it is available regardless of which toolbox you have on the MATLAB path. Only Cartesian-coordinate geometry blocks are supported (not Z-matrix input).

| Field     | Type   | Description                                                            |
|-----------|--------|------------------------------------------------------------------------|
| .symbols  | cell   | Atomic symbols                                                         |
| .xyz      | double | Starting coordinates in Å (X Y Z)                                      |
| .Natoms   | int    | Number of atoms                                                        |
| .charge   | int    | Total molecular charge                                                 |
| .mult     | int    | Spin multiplicity                                                      |
| .title    | char   | Title/comment line(s), joined with a space                             |
| .route    | char   | Full route section, single line                                        |
| .method   | char   | Method parsed from the route (e.g. 'b3lyp'), '' if not found           |
| .basis    | char   | Basis set parsed from the route (e.g. '6-311+g(d,p)'), '' if not found |
| .chk      | char   | %chk link0 value, '' if absent                                         |
| .mem      | char   | %mem link0 value, '' if absent                                         |
| .nproc    | char   | %nprocshared/%nproc link0 value, '' if absent                          |
| .filename | char   | Source file path                                                       |

ginp has the same .symbols/.xyz/.Natoms/ .filename fields as the struct returned by G09\_structure/G16\_structure, so it can be passed directly to G09\_draw\_molecule/G16\_draw\_molecule or G09\_get\_bond\_length/G16\_get\_bond\_length without any conversion.

```
ginp = G_read_input('molecule.gjf');
G16_draw_molecule(ginp, 'ShowAxes', true);
fprintf('%s/%s, charge %d, mult %d\n', ginp.method, ginp.basis, ...
        ginp.charge, ginp.mult);
```

## 9.5 G09\_list / G16\_list

```
G09_list()
T = G09_list()

G16_list()
T = G16_list()
```

Lists every G09\*.m/G16\*.m function found in the toolbox folder, printing the function name and its H1 description line (sorted alphabetically). Also returns the same information as a table with columns .Name, .Description, .File.

```
G09_list();
T = G16_list();
T(contains(T.Description, 'dipole', 'IgnoreCase', true), :)
```



## 10 One-call data extraction

### 10.1 G09\_read\_all / G16\_read\_all

```
T = G09_read_all(filename)
T = G16_read_all(filename)
```

Runs the full set of G09\_XXX/G16\_XXX parsers on a single output file and collects the results into one struct T. The file is read from disk once and the resulting lines are passed to every sub-parser via their 'Lines' option, instead of each sub-parser independently re-reading and re-splitting the file.

| Field             | Source function               | Description                                          |
|-------------------|-------------------------------|------------------------------------------------------|
| .charge           | G09/G16_charges               | Mulliken and APT atomic charges (plotting disabled)  |
| .energy           | G09/G16_energy                | SCF energy and thermochemistry                       |
| .structure        | G09/G16_structure             | Optimized molecular structure                        |
| .dipolar          | G09/G16_dipole_polarizability | Dipole moment and polarisability                     |
| .nmodes           | G09/G16_nmodes                | IR/Raman vibrational normal modes, if present        |
| .spectra          | G09/G16_spectra               | Frequencies, IR/Raman intensities, simulated spectra |
| .route            | G16_route (G16 only)          | Gaussian route section details                       |
| .chargemol.charge | G16_charge_mult               | Total molecular charge, spin multiplicity            |
| .chargemol.mol    | (G16 only)                    |                                                      |

```
T = G16_read_all('zeatin.out');
disp(T.energy)
G16_draw_molecule(T.structure);
```

**Known discrepancy:** the current G09\_read\_all does *not* populate .route or .chargemol, unlike G16\_read\_all, which retains both. If parity with G16\_read\_all is required, add

```
T.route = G09_route(filename, 'Lines', lines);
[c, m] = G09_charge_mult(filename, 'Lines', lines);
T.chargemol.charge = c;
T.chargemol.mol = m;
```

to G09\_read\_all.m.

## 11 Report generation

### 11.1 G09\_write\_report / G16\_write\_report

```
outfile = G09_write_report(T)
outfile = G09_write_report(T, outfile)

outfile = G16_write_report(T)
outfile = G16_write_report(T, outfile)
```

Writes a human-readable text report from a `G09_read_all`/`G16_read_all` struct `T`, formatting a summary of every field present in `T` into a plain `.txt` file. If `outfile` is omitted, the file is named after the source Gaussian file (e.g. `'zeatin.out' → 'zeatin_report.txt'`) in the current folder.

| Argument             | Default                                | Description                                                              |
|----------------------|----------------------------------------|--------------------------------------------------------------------------|
| <code>T</code>       | (required)                             | Struct returned by <code>G09_read_all</code> / <code>G16_read_all</code> |
| <code>outfile</code> | <code>&lt;source&gt;_report.txt</code> | Output <code>.txt</code> path                                            |

Returns the path written to. Sections are included only if the corresponding field is present in `T`, in this order: route, charge/multiplicity, molecular structure (full Cartesian coordinate table), energetics (with thermochemistry if available), dipole moment and polarisability (including the dynamic polarisability table, if present), atomic charges (full per-atom table, sum of charges, dipole-from-charges overlay if computed), vibrational normal modes (frequency/IR/Raman/reduced mass/symmetry table), and a summary of the simulated IR/Raman spectra.

The full broadened-spectrum continuum arrays (`T.spectra.x`, `.IR_cont`, `.Raman_cont`) are deliberately *not* dumped into the report — only the grid range, FWHM, and whether Raman data is present are summarised. Access those arrays directly from `T.spectra` for plotting or export.

```
T = G16_read_all('zeatin.out');
G16_write_report(T);                % -> zeatin_report.txt
G16_write_report(T, 'zeatin_summary.txt'); % explicit output path
```

No differences between the G09 and G16 versions beyond the input struct's origin and the report header ("Gaussian 09"/"Gaussian 16 Calculation Report").

See also `G09_read_all`/`G16_read_all`.

`G16parser` is a Python 3 port of the G16 MATLAB toolbox, covering the same parsing, checkpoint-analysis, and static-visualisation functionality described in the previous sections. It is installable as a standard editable package and exposes one `g16_xxx` function per MATLAB `G16_xxx` counterpart, each returning a `Struct` object (attribute access, e.g. `mol.xyz`, `ch.dipole.Debye`) instead of a MATLAB struct.

```
cd G16parser
pip install -e .
```

```
import G16parser as g16

mol = g16.g16_structure('molecule.out')
g16.g16_draw_molecule(mol, show_axes=True)

ch = g16.g16_charges('molecule.out', show_dipole=True)
```

```
oe = g16.g16_orbital_energies('molecule.out')
g16.g16_draw_orbital(oe)

T = g16.g16_read_all('molecule.out')    # everything in one call,
    single file read
```

**Requirements:** Python  $\geq 3.9$ , numpy, pandas, matplotlib (installed automatically via `pip install -e .`). No Gaussian installation or compiled dependency is needed.

## 11.2 Function reference

Every function mirrors its MATLAB `G16_xxx` counterpart in name, parameters (in `snake_case`), defaults, and returned fields — see the corresponding MATLAB subsection earlier in this manual for full parameter/field documentation. All extraction functions accept an optional `lines=` parameter (pre-read file lines) mirroring the MATLAB `'Lines'` option; `g16_read_all` uses this internally to read the file exactly once.

| Function                          | Description                                                                                                                                                                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>g16_read_input</code>       | Reads a Gaussian <i>input</i> file ( <code>.gjf/.com/.in</code> ): link0 commands, route, title, charge/multiplicity, and starting geometry — output <code>Struct</code> is compatible with <code>g16_draw_molecule/g16_get_bond_length</code> |
| <code>g16_structure</code>        | Molecular geometry (symbols, xyz as <code>np.ndarray</code> , atom count)                                                                                                                                                                      |
| <code>g16_energy</code>           | SCF energy and thermochemistry                                                                                                                                                                                                                 |
| <code>g16_convergence</code>      | Geometry-optimisation convergence criteria per step ( <code>plot=True</code> for a matplotlib figure)                                                                                                                                          |
| <code>g16_charges</code>          | Mulliken/APT atomic charges, optional dipole overlay and 3D plot                                                                                                                                                                               |
| <code>g16_dipole_polar</code>     | Dipole moment and (static/dynamic) polarisability                                                                                                                                                                                              |
| <code>g16_nmodes</code>           | Vibrational normal-mode displacement vectors                                                                                                                                                                                                   |
| <code>g16_spectra</code>          | IR/Raman spectra (stick + Lorentzian-broadened continuum)                                                                                                                                                                                      |
| <code>g16_orbital_energies</code> | HOMO/LUMO and the full occupied/virtual MO spectrum                                                                                                                                                                                            |
| <code>g16_charge_mult</code>      | Molecular charge and spin multiplicity                                                                                                                                                                                                         |
| <code>g16_route</code>            | Route section string                                                                                                                                                                                                                           |
| <code>g16_get_bond_length</code>  | Bond-length table (pandas DataFrame) from covalent radii                                                                                                                                                                                       |
| <code>g16_gaussian_version</code> | Detects the Gaussian version/revision (works on <code>.fchk</code> via a sibling <code>.log/.out</code> )                                                                                                                                      |
| <code>g16_hyperpolar</code>       | Dipole hyperpolarisability ( $\beta$ )                                                                                                                                                                                                         |
| <code>g16_tddft</code>            | TD-DFT excited states                                                                                                                                                                                                                          |
| <code>g16_read_all</code>         | Runs the full extraction set in one call, reading the file only once                                                                                                                                                                           |
| <code>g16_draw_molecule</code>    | 3D CPK ball-and-stick render (matplotlib <code>mplot3d</code> )                                                                                                                                                                                |
| <code>g16_draw_mode</code>        | 3D structure with a vibrational mode's displacement arrows                                                                                                                                                                                     |
| <code>g16_draw_orbital</code>     | Orbital energy-level diagram with HOMO–LUMO gap arrow                                                                                                                                                                                          |
| <code>g16_animate_mode</code>     | Exports an MP4 animation of a vibrational mode (requires <code>ffmpeg</code> on PATH)                                                                                                                                                          |
| <code>g16_mode_viewer</code>      | Interactive Tkinter window to browse/render vibrational modes, sortable by mode number, IR, or Raman intensity, with an "Animate mode (MP4)..." button                                                                                         |
| <code>g16_list</code>             | Lists every function in the toolbox with its one-line description (pandas DataFrame)                                                                                                                                                           |
| <code>g16.write_report</code>     | Writes a human-readable <code>.txt</code> summary report from a <code>g16_read_all Struct</code>                                                                                                                                               |

### 11.3 Selected signatures

```

g16_read_input(filename)
g16_structure(filename, orientation="auto", step="last", lines=None)
g16_energy(filename, step="last", lines=None)
g16_convergence(filename, plot=False, lines=None)
g16_nmodes(filename, section="last", modes=None, lines=None)
g16_spectra(filename, FWHM=10, xmin=0, xmax=4000, dx=1, normalize=
    False,
                plot=False, section="last", lines=None)
g16_charges(filename, type="Mulliken", mode="atom", plot=True,

```

```

        atom_scale=0.35, bond_tol=1.30, font_size=8, color_scale="
            RdBu",
        threshold=0.0, show_dipole=False, dipole_origin="negcharge
            ",
        dipole_scale=1.0, dipole_color=(0, 0.6, 0),
            dipole_line_width=2.5,
        show_dipole_label=True, dipole_font_size=11, dipole_units
            ="Debye",
        lines=None)
g16_draw_molecule(mol, atom_scale=0.35, bond_tol=1.30, show_labels=
    True,
                    show_legend=True, title="", bg_color=(0.95, 0.95,
                        0.95),
                    ax=None, show_axes=False, axes_length=None,
                        bond_list=None)
g16_animate_mode(mol, nm, mode_idx, filename=None, scale=1.5,
    flip_sign=False,
                    atom_scale=0.35, bond_tol=1.30, show_labels=False,
                        frames_per_cycle=30, n_cycles=2, fps=20, view=None)
g16_mode_viewer(filename, scale=1.5, atom_scale=0.35, bond_tol=1.30,
    show_labels=False, flip_sign=False, arrow_color=None,
        font_size=10, **extra_opts)
g16_get_bond_length(mol, tolerance=1.15, include_h=True,
    sort_by="distance", save_as="")
g16_hyperpolar(filename, units="au", lines=None)
g16_tddft(filename, nstates=math.inf, plot=False, fwhm_ev=0.30, lines=
    None)
g16_write_report(T, outfile=None)

```

## 11.4 Notes on the port

**3D interactivity.** matplotlib's `mplot3d` has no MATLAB-style `rotate3d` widget; use the interactive backend's normal mouse/toolbar controls instead.

**Naming exception: `g16_read_input`.** On the MATLAB side this function has no version prefix at all (`G_read_input`, identical copies in both the `G09/` and `G16/` folders) because the Gaussian input file format does not differ between versions. Since only a `G16parser` Python package exists (there is no separate `G09parser`), the port simply follows this package's `g16_` naming convention like every other function here.

**Backend selection.** The package forces the `TkAgg` matplotlib backend (if `Tk` is available and `pyplot` has not been imported yet) instead of the native macOS backend: on some older matplotlib releases, interactively rotating a 3D plot with the native Cocoa backend can segfault the whole Python process. Call `matplotlib.use(...)` yourself *before* importing `G16parser` if a different backend is needed.

**Bug found during porting, MATLAB-side only:** in the original `G16_tddft.m`, MATLAB's `regexp` silently drops the optional trailing `<S**2>=` capture group (a MATLAB-specific regex-engine quirk), so `td.S2` is always `NaN` in MATLAB even when the tag is present in the file. This Python port's `re`-based parser does not have that bug and extracts `S2` correctly. Consider this a known discrepancy between the two implementations rather

than a porting error.

**g16\_mode\_viewer** is a Tkinter rewrite of `G16_modeViewer.m`'s `uifigure` GUI, with the same controls (mode selector, order-by, per-option redraw, title, save-as PDF/EPS/JPEG, and an "Animate mode (MP4)..." button wired to `g16_animate_mode`, using the current option values and the displayed mode figure's camera orientation). One simplification: "target figure" always means the most recently drawn mode window, not whichever window the user last clicked on (MATLAB's `groot().CurrentFigure` has no simple Tkinter equivalent). In an IPython/Spyder console with the Tk graphics backend active, the viewer detects the running event loop and returns to the prompt immediately instead of blocking; run as a plain script, it blocks until the viewer window is closed.

**Known issue:** `g16_mode_viewer` can segfault when run from *inside* Spyder's IPython console (a crash inside ipykernel's own periodic Tk event-loop pump, not in this toolbox's code) — confirmed working correctly when run from a plain terminal/script instead:

```
python3 -c "import G16parser as g16; g16.g16_mode_viewer('file.out
  ')"
```

All other functions, including static plots, work fine inside Spyder.

**g16\_animate\_mode** mirrors `G09_animate_mode.m`/`G16_animate_mode.m`: it oscillates the molecule along the mode's displacement vector and saves an MP4 via `matplotlib.animation.FuncAnimation` with `FFMpegWriter`. **Requires ffmpeg installed and on PATH** (brew install ffmpeg on macOS, `sudo apt install ffmpeg` on Ubuntu/Debian) — `matplotlib` does not bundle a video encoder itself; without it the call fails with `FileNotFoundError: ... 'ffmpeg' at the final encoding step` (frame generation itself does not need `ffmpeg`). Bonds are computed once from the equilibrium geometry via `g16_get_bond_length` and passed to `g16_draw_molecule` through its `bond_list` parameter, so they stay fixed across all frames instead of flickering as instantaneous distances cross `bond_tol`. By default the animation uses `matplotlib`'s default 3D view; pass `view=(ax.azim, ax.elev)` to start from an already-rotated figure's orientation instead.

**g16\_write\_report** takes the `Struct` returned by `g16_read_all` and writes the same section-by-section plain-text report as `G09_write_report.m`/`G16_write_report.m` (route, charge/multiplicity, structure, energetics, dipole/polarisability, atomic charges, normal modes, spectra summary), using Python `getattr`-based field checks so that sections are included only when present, exactly mirroring the MATLAB `isfield` logic.

**Bond order (single/double/triple).** `g16_draw_molecule` mirrors `G09_draw_molecule.m`/`G16_draw_molecule.m`: for C-C, C-N, and C-O pairs, bond order is estimated purely from bond length (any other element pair is always drawn as a single bond) and rendered as 1/2/3 parallel lines. This is a geometric estimate, not Gaussian's own bond-order analysis (e.g. Wiberg/NBO indices). The `bond_list` parameter accepts an optional 3rd column with a pre-computed order, used by `g16_animate_mode` (via `g16_get_bond_length`) to keep both bond topology and bond order fixed across all animation frames. The C-C double/single threshold is set to 1.36 Å rather than the generic reference-length midpoint, so symmetric aromatic rings (C-C  $\approx$  1.39–1.40 Å, no real length alternation) render as all-single instead of all-double — see the MATLAB `G09_draw_molecule`/`G16_draw_molecule` subsection earlier in this manual for the full

rationale.

## 12 Publication and licensing

### 12.1 Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Claude (Anthropic) to assist with code drafting, refactoring, and documentation of the G09/G16 toolbox and its associated user manual. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

### 12.2 Note on the Use of AI-Assisted Development Tools

Portions of the codebase and documentation described in this manual — including the G09\_/G16\_ MATLAB function pairs, the `g16toolbox` Python package, the `G09_modeViewer.m`/`G16_modeViewer.m` interactive GUIs, and this reference manual — were developed with the assistance of Claude (Anthropic), an AI-based coding tool, used interactively under the direct supervision of the author. All AI-generated code was tested against reference Gaussian 09/16 output files and reviewed by the author prior to inclusion. Algorithm design, scientific validation, and correctness of results remain the sole responsibility of the author.

### 12.3 Licence

This software is released under the **BSD 3-Clause License**, an OSI-approved open source license accepted by SoftwareX. See the `LICENSE.txt` file distributed with the source code for the full licence text.

### 12.4 Citation

If this toolbox contributes to published research, please acknowledge it by citing the source repository:

S. Trusso, *G09/G16 Toolbox: MATLAB and Python libraries for parsing, analysing and visualising Gaussian 09/16 output files*, [https://github.com/nello62/Gaussian09G16\\_output\\_Matlab\\_parser](https://github.com/nello62/Gaussian09G16_output_Matlab_parser) (accessed July 23, 2026).